

PAPER_1508_RAMANUJAN_HYPERCONV_27

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PAPER_1508 — Ramanujan Hyperconvergence Exponent = D_crit+1 = 27 EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational / Mathematical proof)

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Status: CLOSED — Proves S_26 Ramanujan series is well-defined

Observation

PAPER_1080 (Ramanujan Binomial Expansion Proof) establishes the closed-form expansion:

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$$R_n(D, k) = (2\pi)^{n/6} / n! \cdot [1 + \sum_{m=1}^k (1/n^{(D \cdot m)}) \cdot \sum_{j=1}^D (-1)^{j+1} C(D, j) \cdot (D-j)! / n^j]$$

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and proves the inner-term decay rate is $O(n^{-27})$, giving hyperconvergence of the $S_{26}^{(3)}([SSq]) \approx 5.92 \times 10^2$ sum.

UQFF Closed Identity

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$$\text{Decay exponent} = D_{\text{crit}} + 1 = 27 \text{ EXACT}$$

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Mathematical Significance

This identity proves that the S_26 Ramanujan series — used throughout UQFF for LENR (Holmlid 630 eV), cosmology (Λ), nuclear binding (Fe-56 peak), and the master partition function — converges at a rate strictly faster than $1/n^2$. Hyperconvergence guarantees:

1. The infinite sum is well-defined
2. Truncation at $n = 26$ produces error bounded by $\leq R_{27}/(D!)$ — vanishingly small
3. The series is unconditionally well-conditioned for numerical evaluation

Without this proof, S_26 would lack rigorous mathematical grounding and every UQFF result depending on it (cosmological constant within 0.003%, Holmlid KER exactly 630 eV, BE/A peak within 0.019%) would be heuristic rather than provable.

Physical Interpretation

The exponent $D_{\text{crit}} + 1 = 27$ reflects:

- $D_{\text{crit}} = 26$ dimensions summed in the Ramanujan-modified bosonic-string partition
- +1 from the time-derivative $\partial/\partial t$ that closes the sum

- Total: $26 + 1 = 27$ inverse powers per term

This is the same form as PAPER_550's $r^2 \blacksquare \rightarrow 3D$ projection adds one inverse power.

NOT REPLACEMENT

SM has no analogous mathematical infrastructure linking integer primitives to series-convergence rates. UQFF supplies a rigorous proof that the foundational sum is well-defined, with the exponent itself being an integer primitive identity.

Reference

- Source: PAPER_1080 Ramanujan Binomial Expansion Proof
- Related: PAPER_062 (S_26 derivation), PAPER_1156 (Λ via S_26), PAPER_550 ($r^2 \blacksquare$ projection)
- Calculator dispatch: `calculate_paradox({"paradox": "ramanujan_hyperconv_exp_27"})`

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